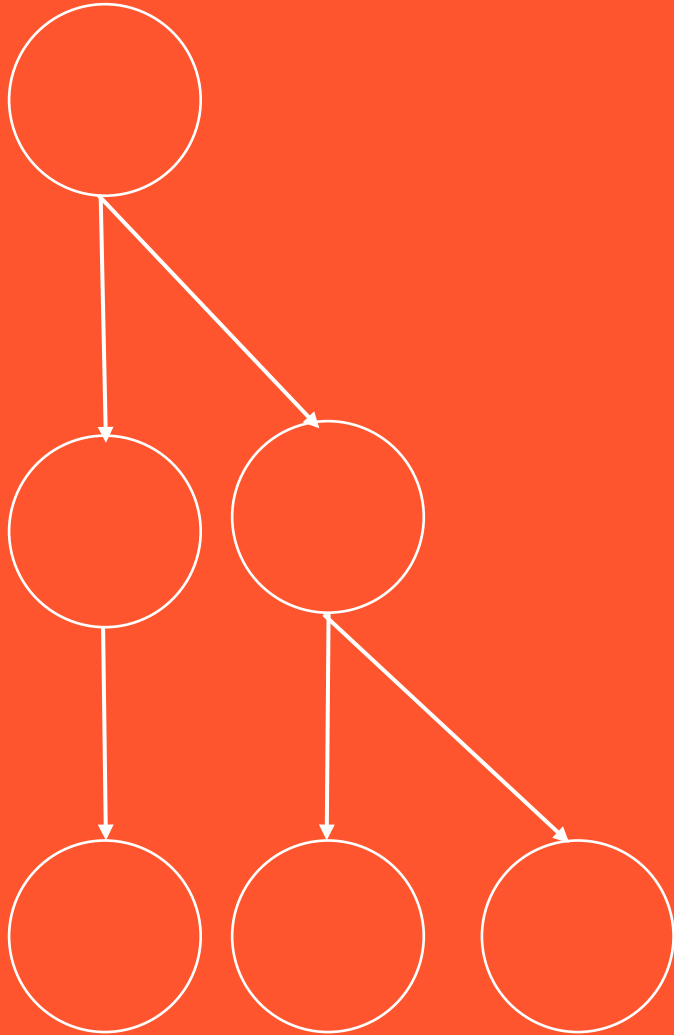


BCOG 100

Language

Lecture 3: The Neuroscience of Language



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 11-3

Password: broca

Question: The article says all languages have a grammar. What does that mean?

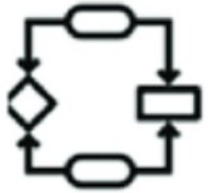


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Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap & Overview

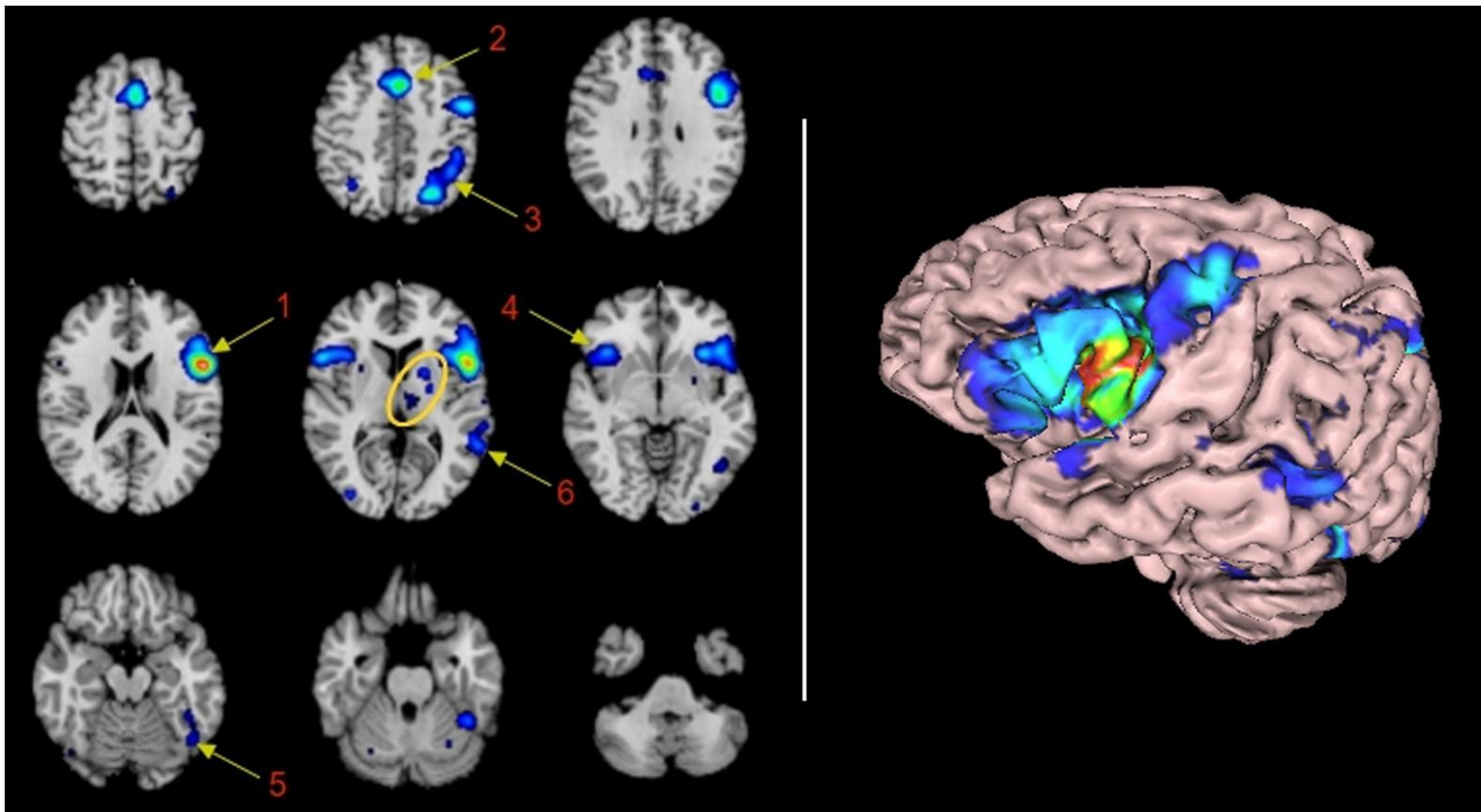
Last Time:

The psychology of language, and the “rules vs. statistics” debate

Today:

From theories of mind to brain systems
What would each theory look like neurally?

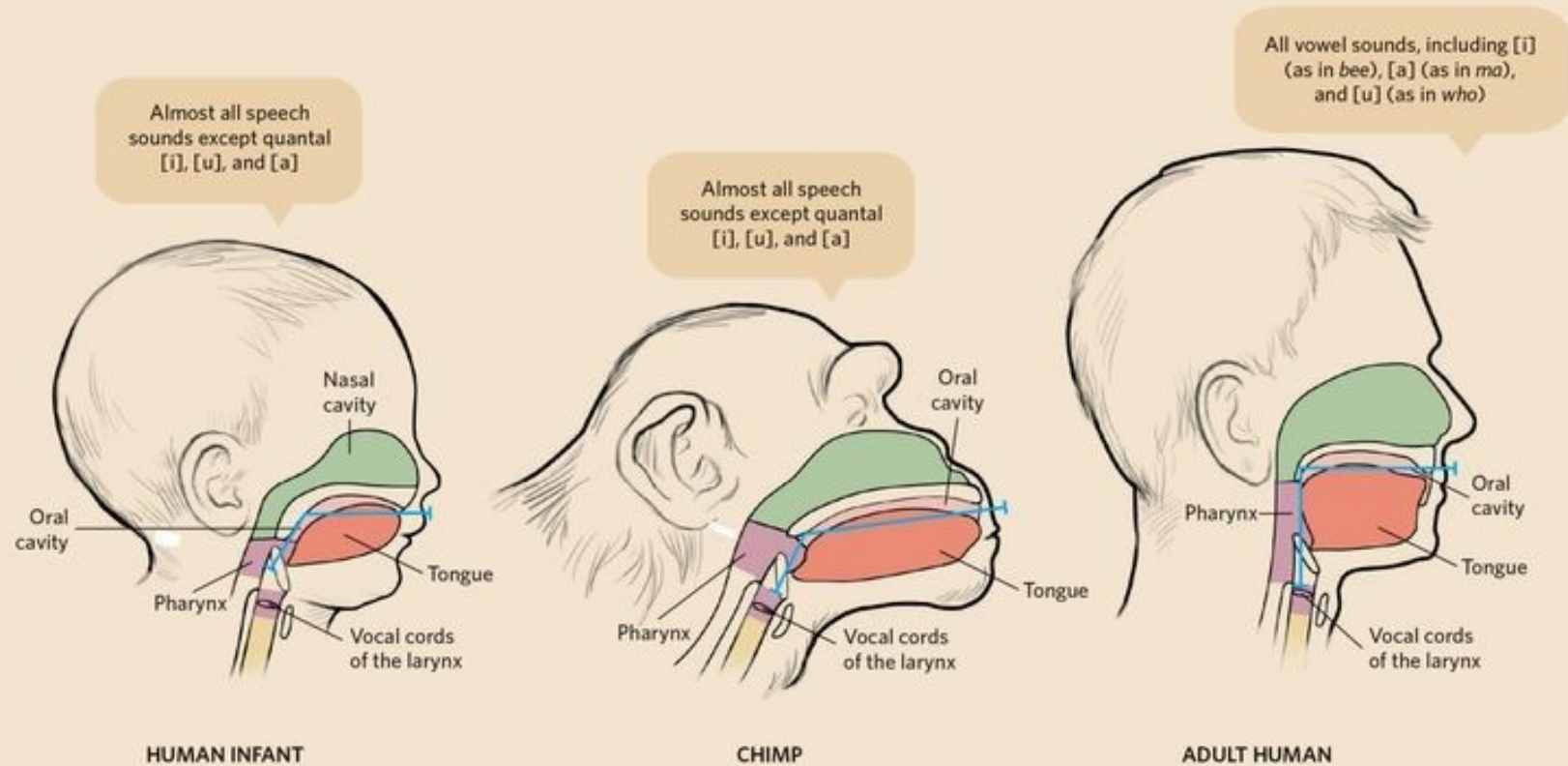
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What Neuroscience Adds, and Doesn't

- Brain data illuminate processes, not philosophies
- Localization is not explanation
- Implementation complements, not replaces, theory

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Evolving the Vocal Tract

- Speech depends on anatomy as well as thought
- Small physical shifts, vast communicative payoffs
- Evolution shaped how sound could carry meaning

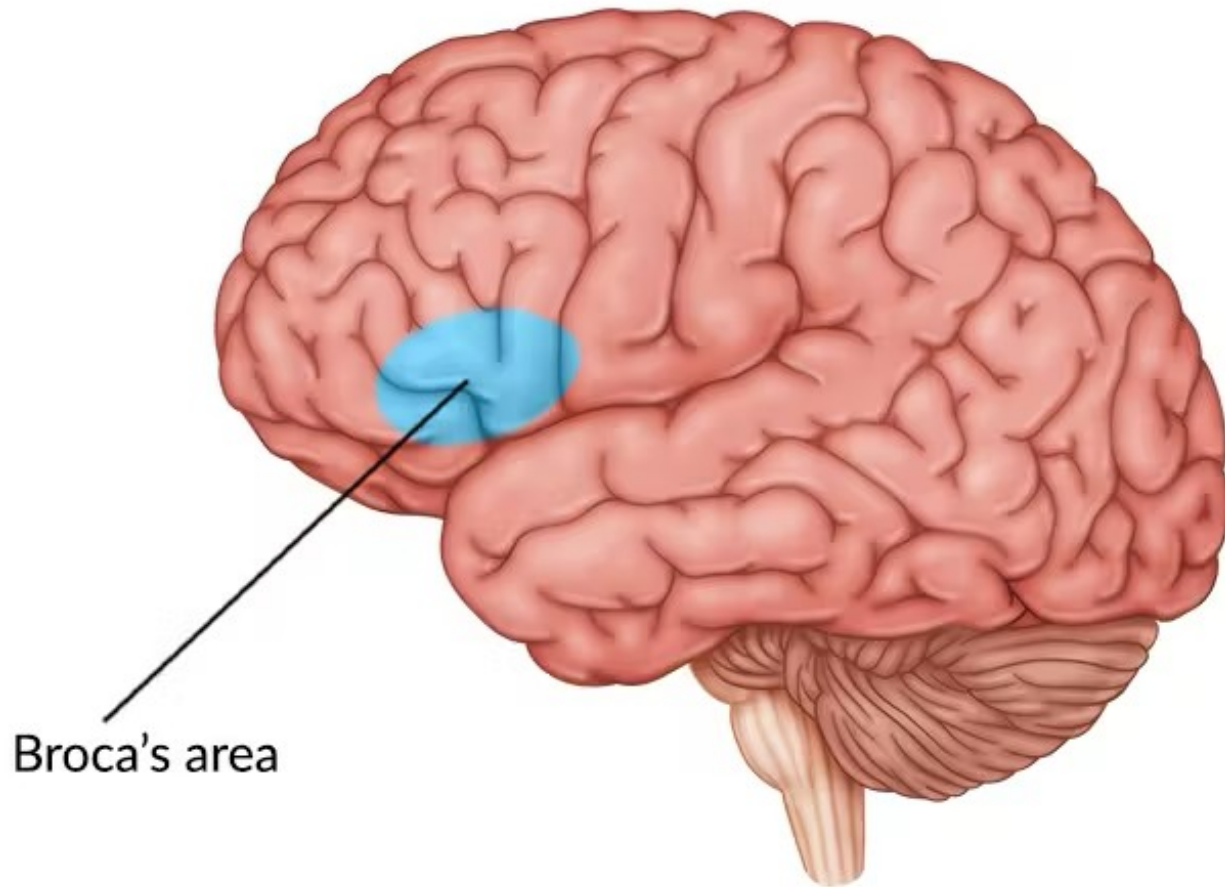
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Motor and Respiratory Control for Speech

- Speaking is skilled movement, not just sound
- Motor control made language fluid and flexible
- The brain learned to choreograph breath, lips, and larynx

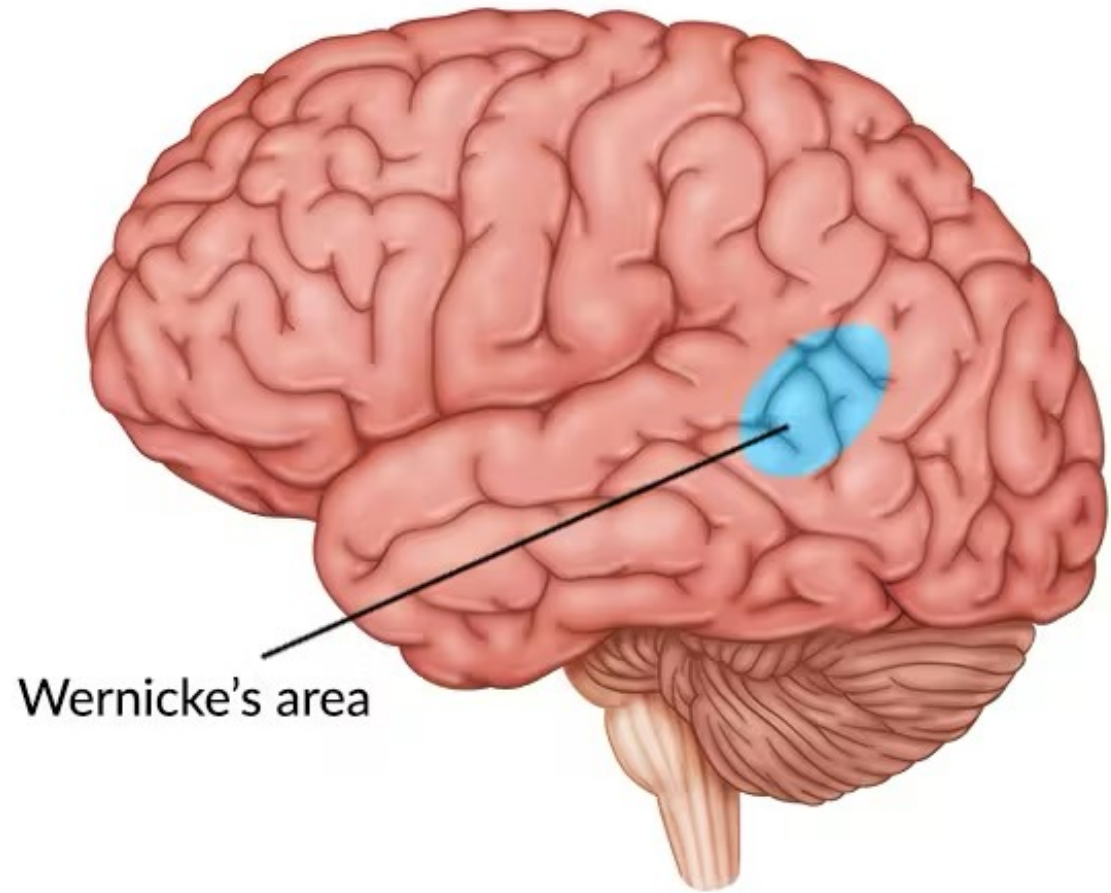
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Broca's Discovery and Expressive Aphasia

- Language and brain meet in a single patient
- Speech lost, comprehension spared
- Broca's area and the birth of localization

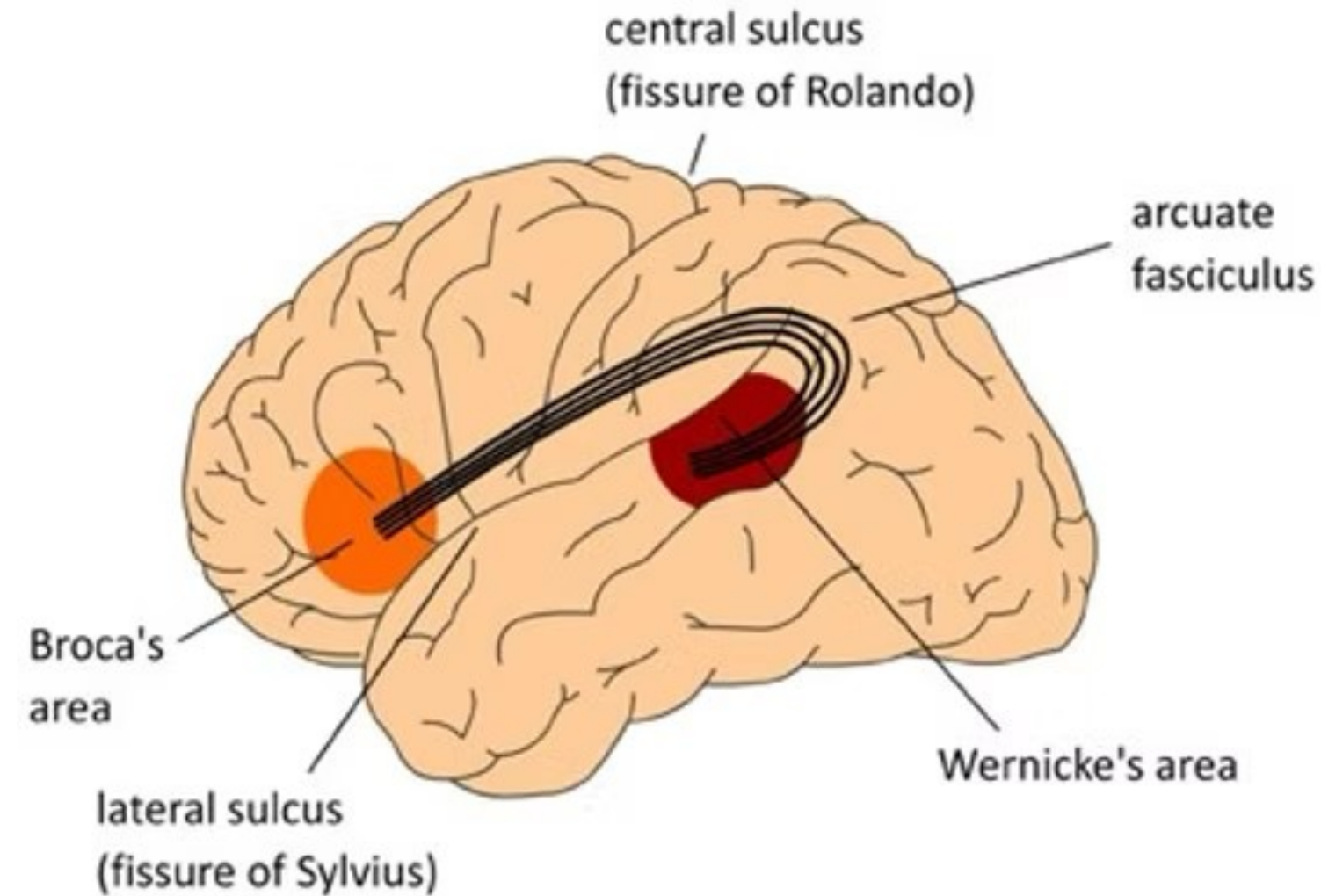
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Wernicke's Discovery and Receptive Aphasia

- Fluent speech, but meaning lost
- Comprehension depends on posterior networks
- Two halves of the classical language model emerge

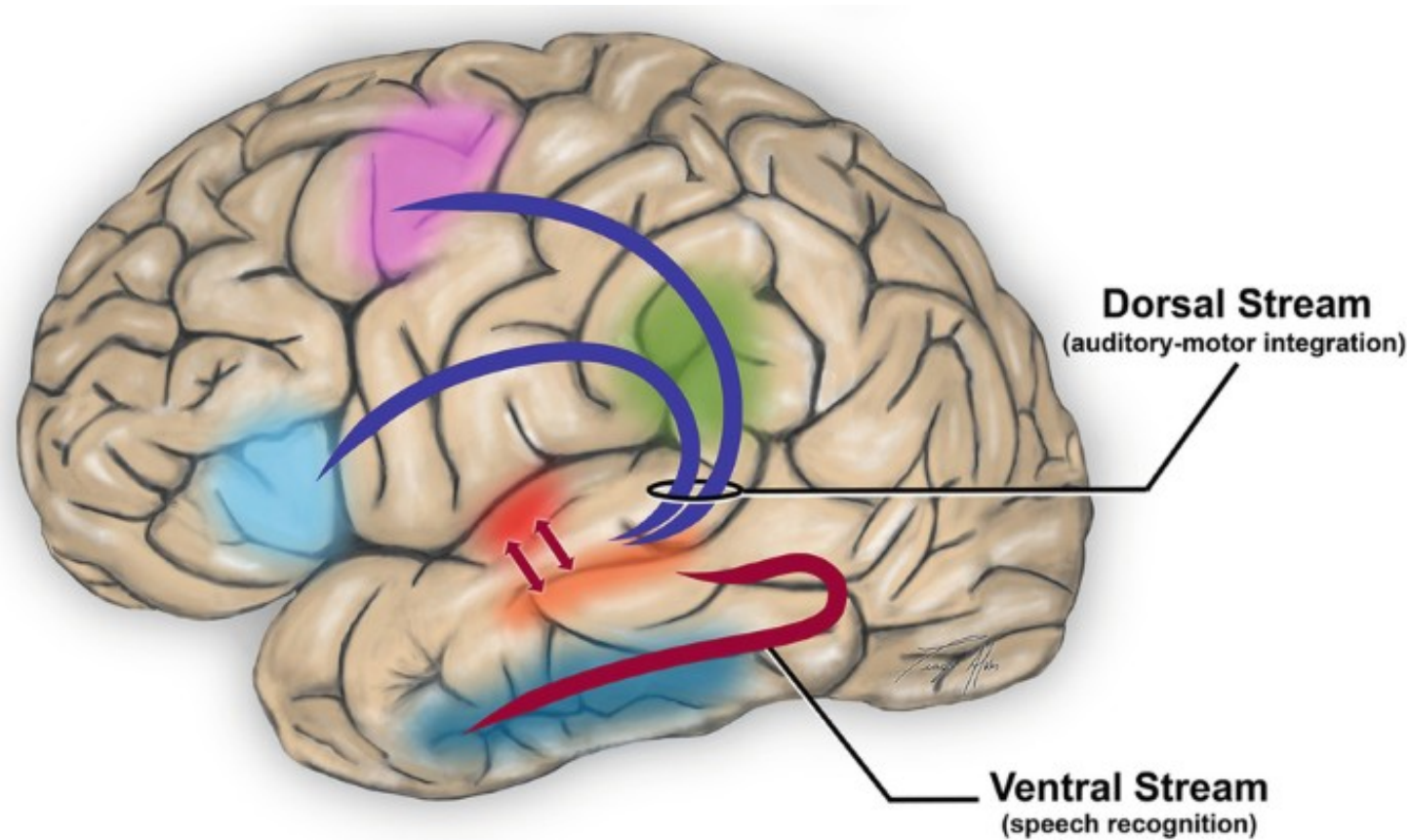
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The Classical Circuit: Broca ↔ Wernicke via Arcuate Fasciculus

- A wiring diagram for language is born
- Production and comprehension linked by a pathway
- Simple, elegant—and incomplete

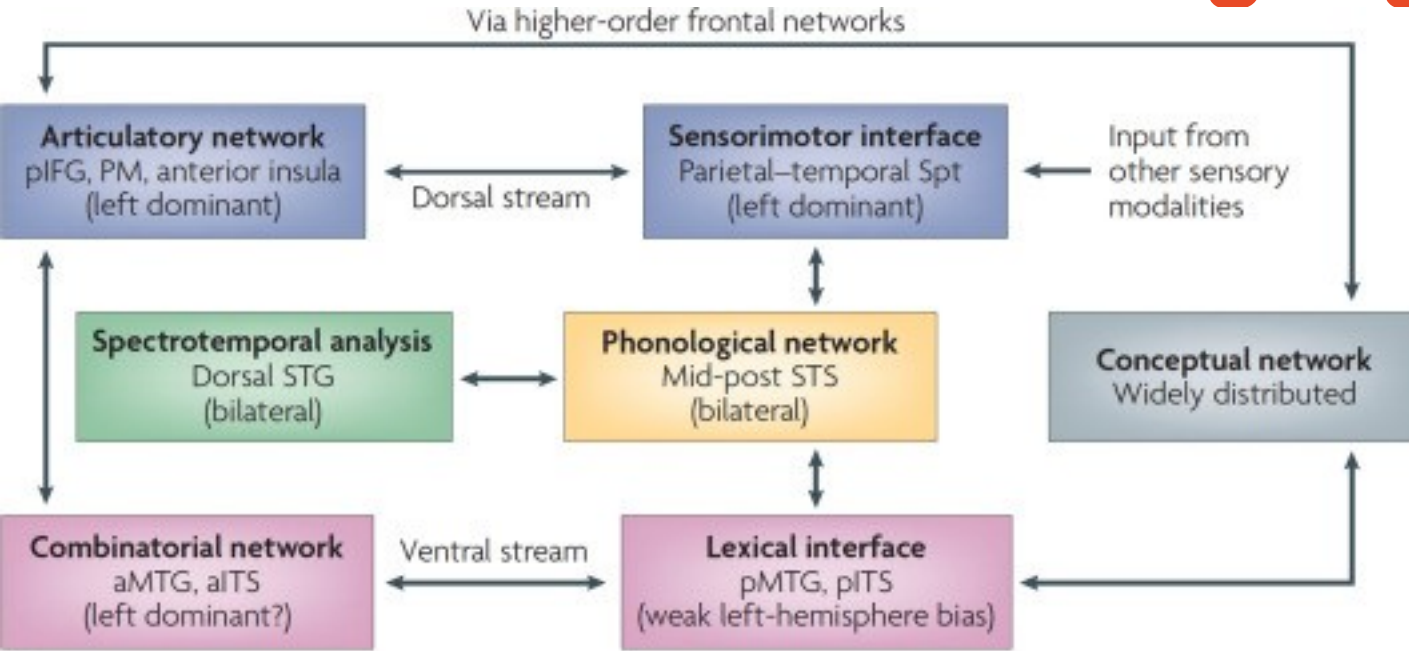
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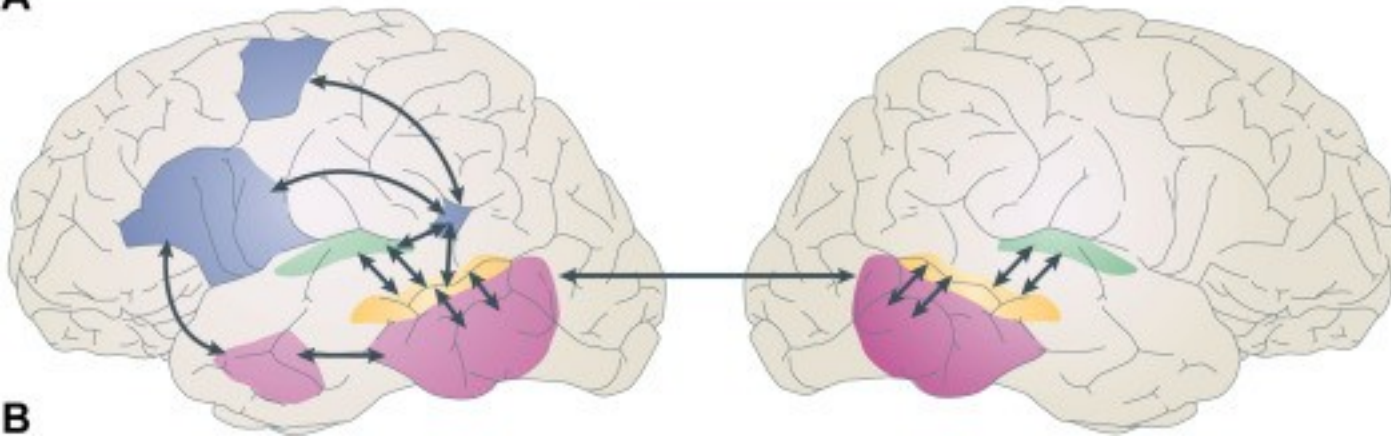
Limits of the Classical Story

- Two centers cannot explain a distributed skill
- Language relies on loops, not lines
- Modern models replace the circuit with interacting streams

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A

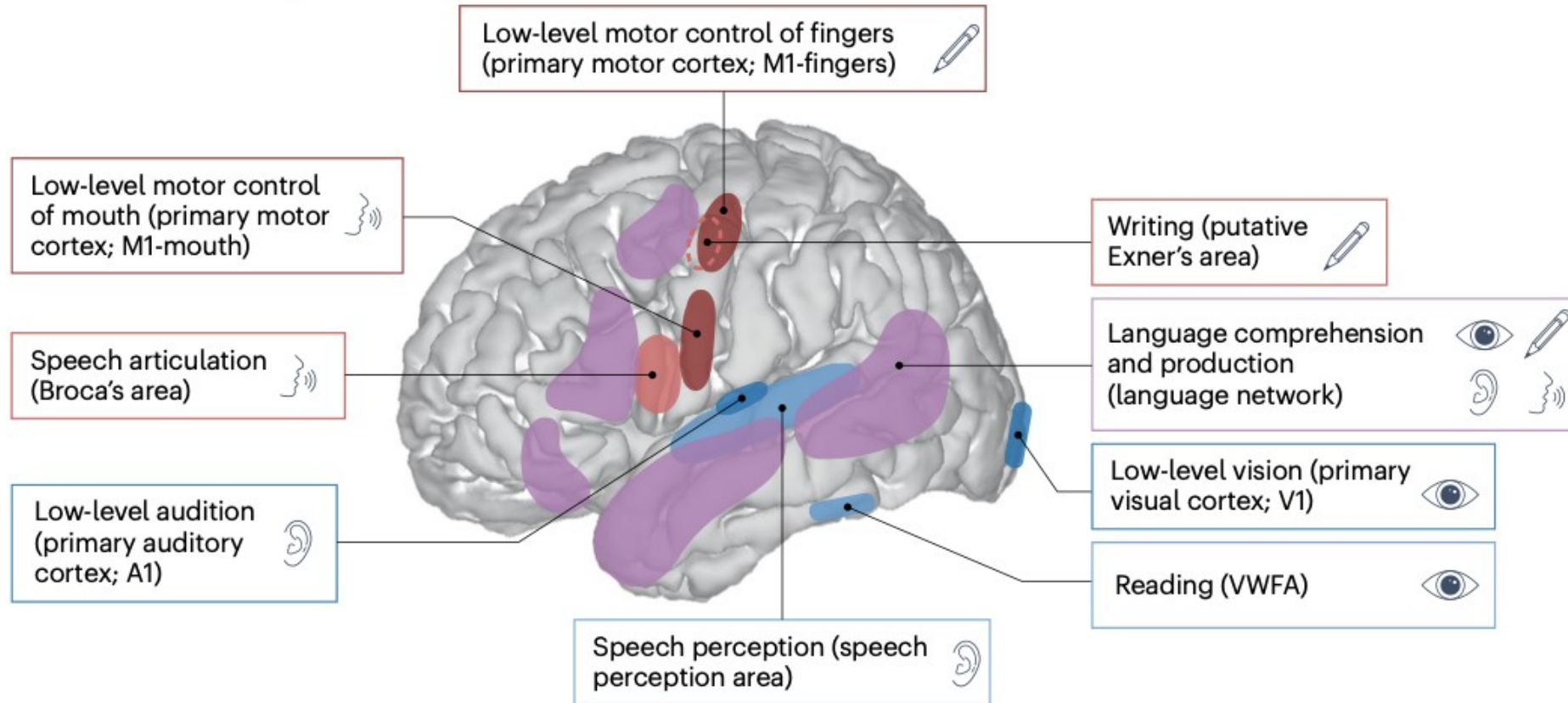


The Dual-Stream Architecture in Detail

- Two interacting pathways, one integrated system
- Dorsal stream: sound to articulation
- Ventral stream: sound to meaning

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a Brain areas that respond to linguistic stimuli



The Core Language Network

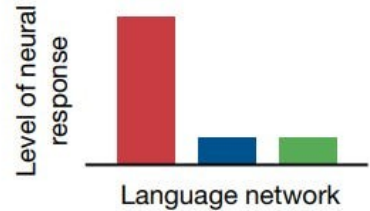
- A functionally defined network for language
- Frontal and temporal regions act together
- Selective for sentences, not just sound or memory

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b Language network

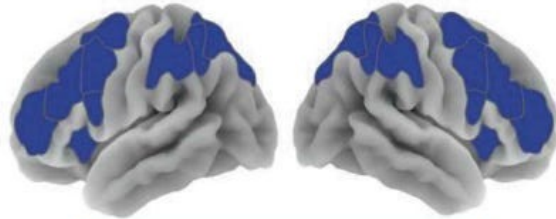


Supported functions:
Language comprehension
Language production

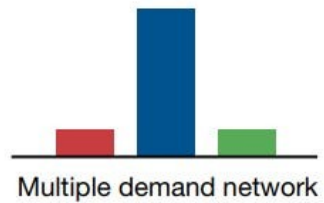


Some networks that support thinking and reasoning

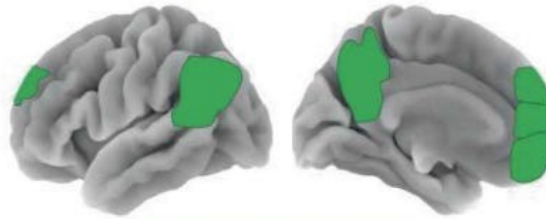
Multiple demand network



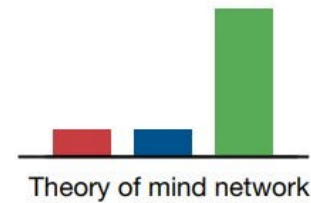
Supported functions:
Executive functions
Novel problem solving
Mathematics
Some forms of reasoning
Computer code comprehension



Theory of mind network



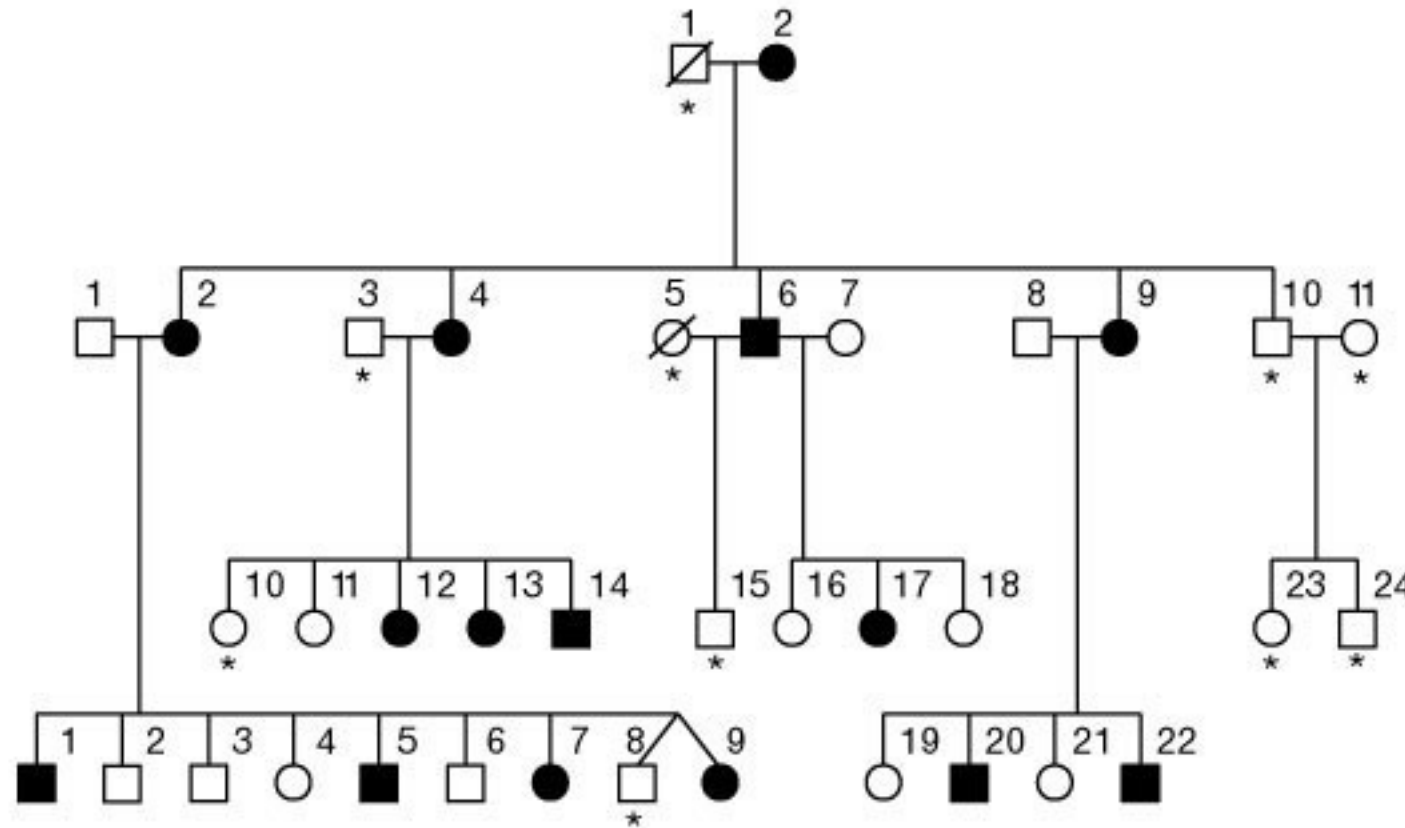
Supported functions:
Social reasoning
Mentalizing



Domain-Specific vs. Domain-General Systems

- Language network, control network, and meaning network
- Selective specialization meets flexible coordination
- The brain balances focus with integration

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Comparative Vocal Learning and FOXP2

- Other species imitate; few can truly learn to speak
- Vocal learning links communication to motor learning
- FOXP2 offers a genetic window—but not a “language gene”

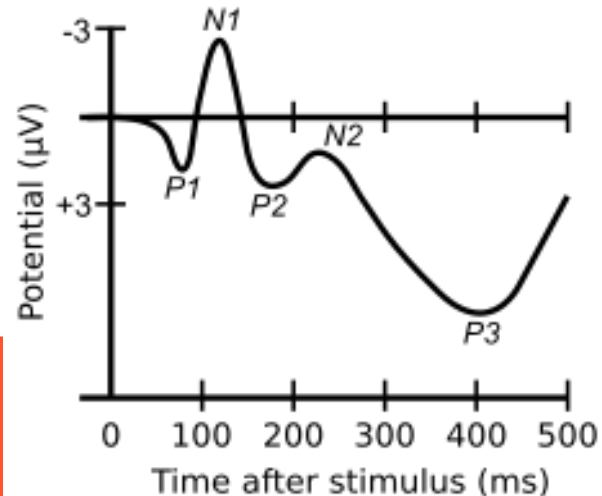
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Electroencephalogram (EEG)

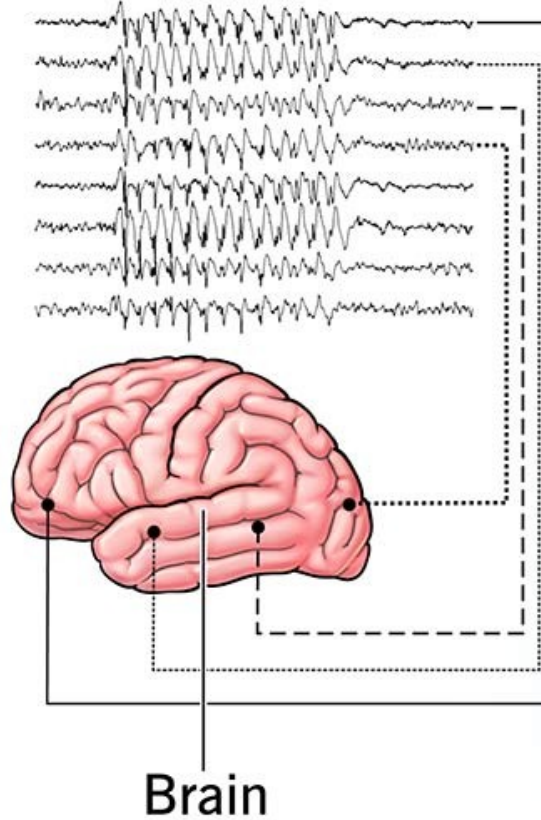
Methods: EEG and ERP

Basics

- Measuring the brain in time, not space
- ERPs track processing millisecond by millisecond
- Revealing how comprehension unfolds in real time



EEG (scan of brainwaves)



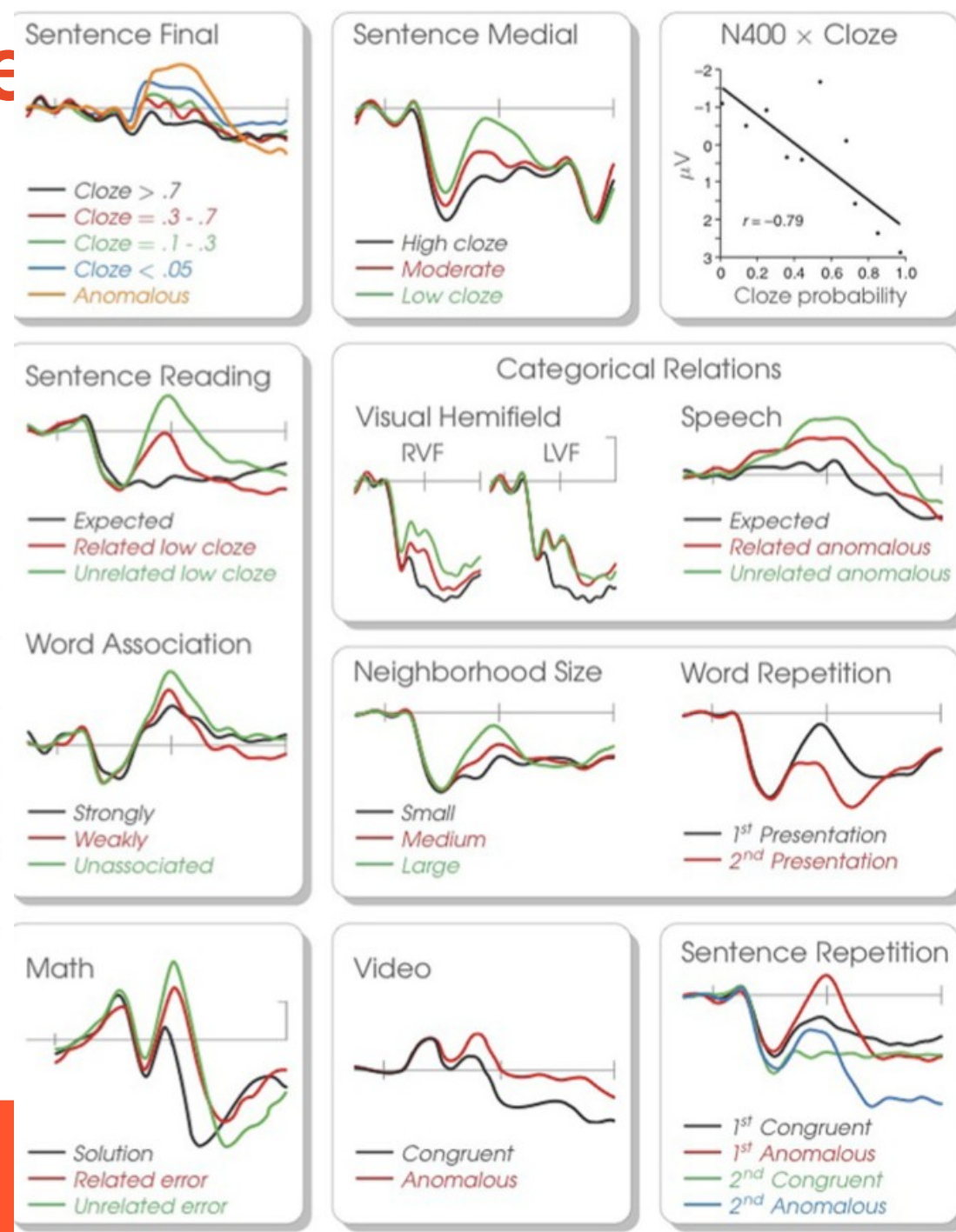
Electrodes
glued to scalp



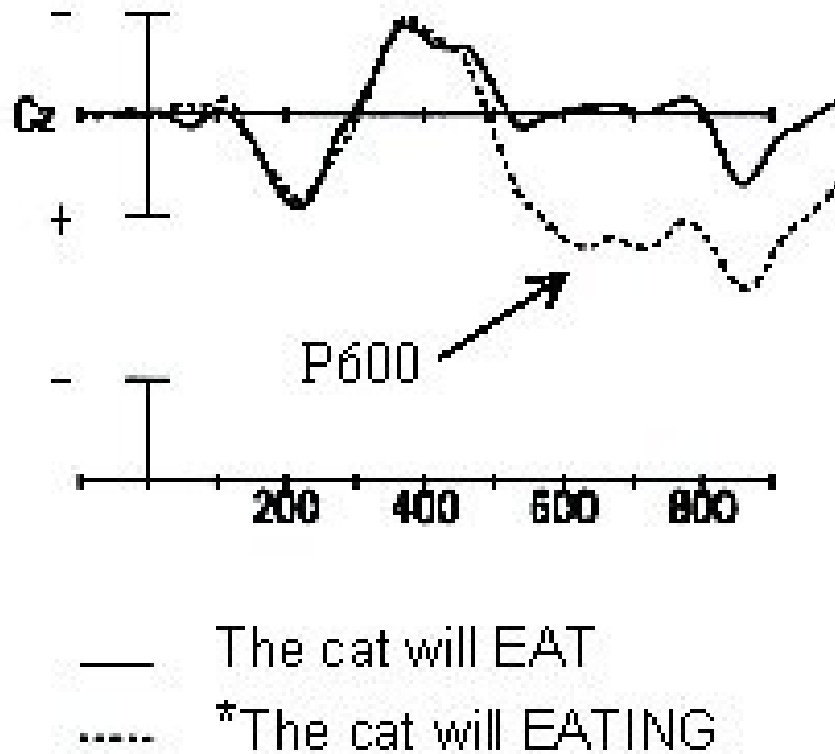
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The N400: Meaning, Memory, and Prediction

- A window into comprehension in real time
- Prediction and meaning draw on memory
- The brain prepares before words arrive



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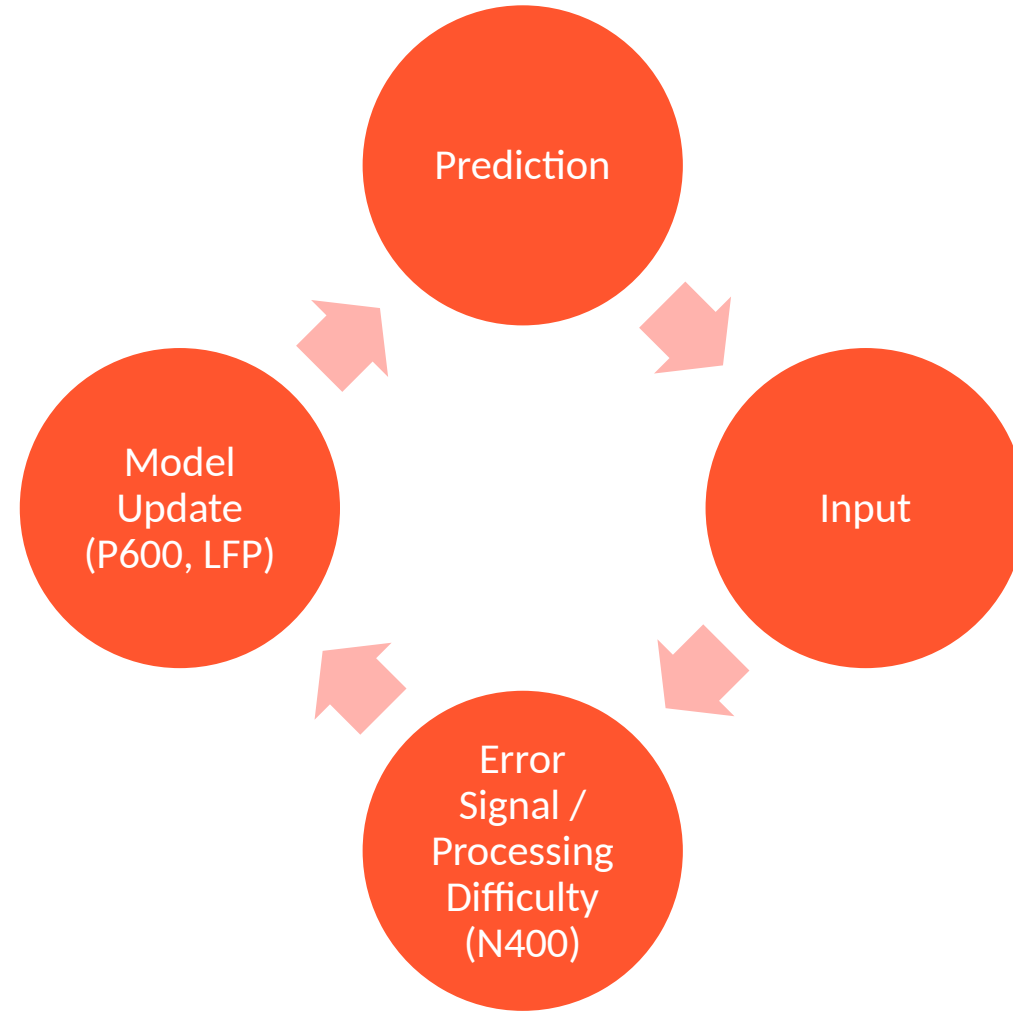
The P600: Reanalysis and Adaptation

- A signal of surprise and repair
- Syntax, semantics, or conflict?
- The brain's model-updating response

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Integrating the Signals: Prediction and Reanalysis

- Comprehension is continuous, not discrete
- The brain anticipates, evaluates, and revises
- Prediction and repair form one adaptive loop



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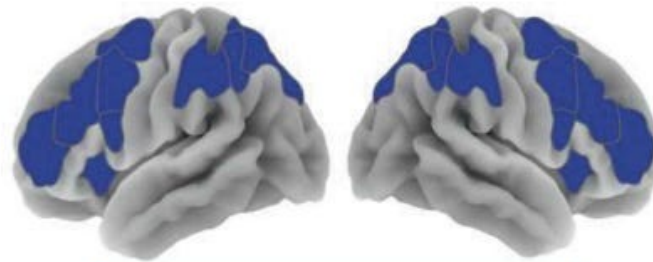
b Language network



Supported functions:
Language comprehension
Language production

Some networks that support thinking and reasoning

Multiple demand network



Supported functions:
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Novel problem solving
Mathematics
Some forms of reasoning
Computer code comprehension

Theory of mind network



Supported functions:
Social reasoning
Mentalizing

Bilingualism and the Language Brain

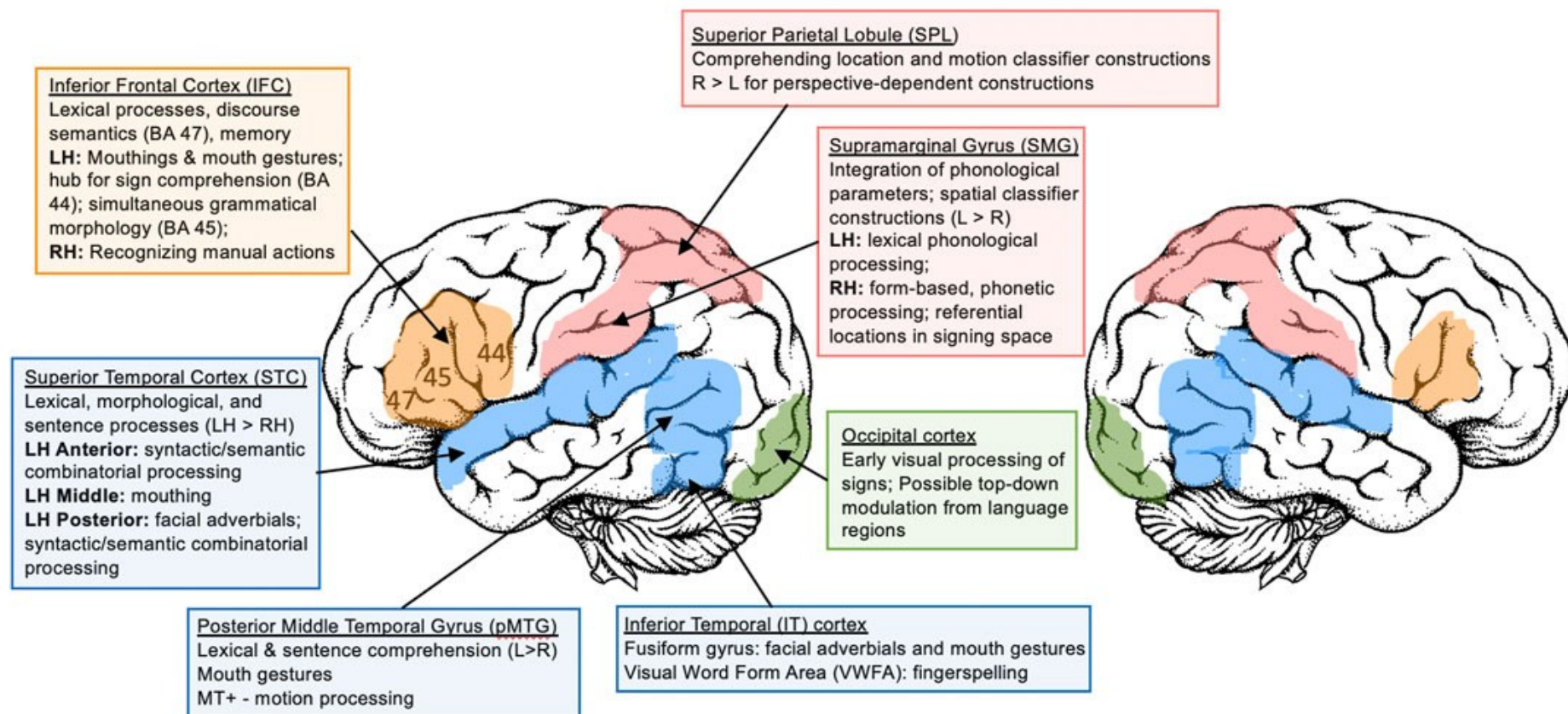
- Two languages, one network
- Shared circuits with dynamic control
- Flexibility without duplication

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Signed Languages and Modality Independence

- Different channels, same architecture
- Structure over surface
- Language transcends speech

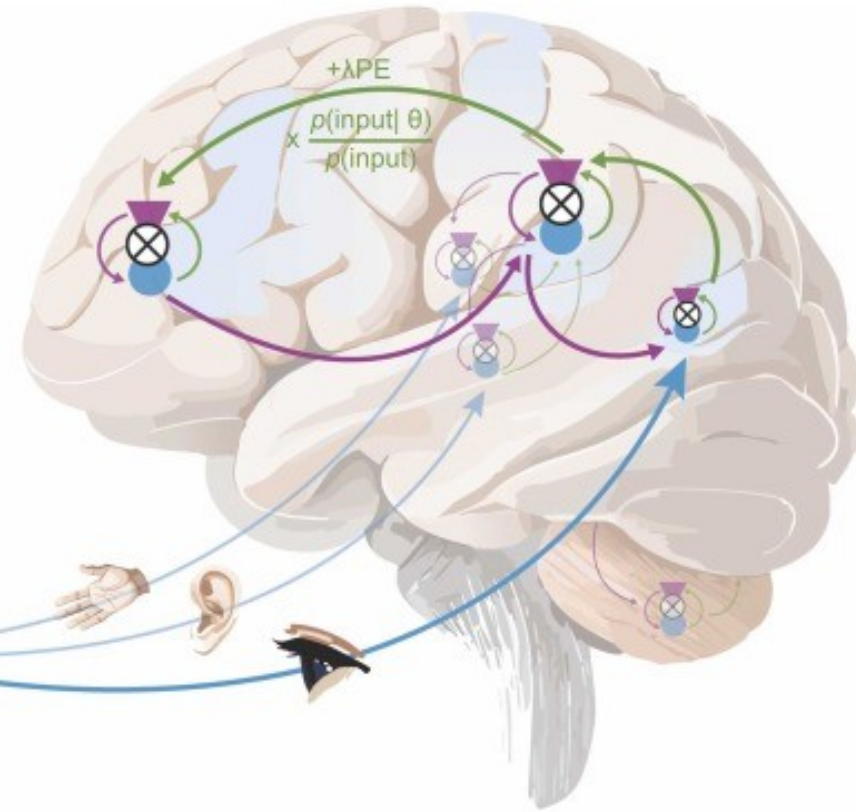
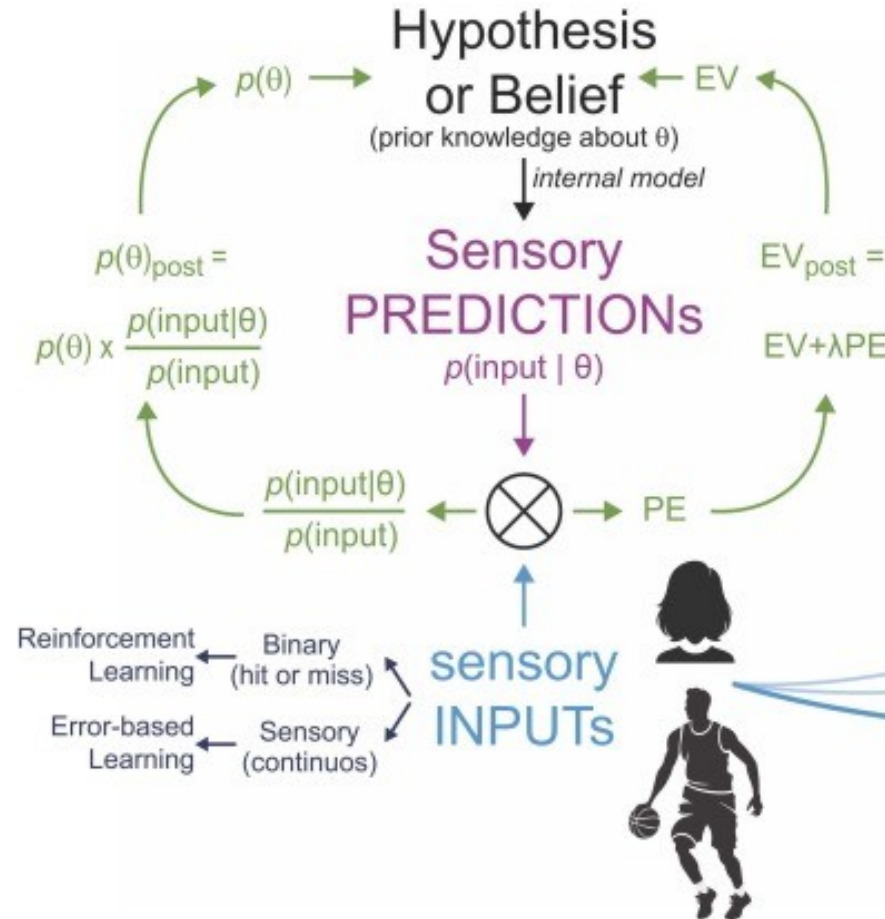
Sketch of the neural network for sign comprehension



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Predictive Coding and the Evolution of Inner Simulation

- From movement to mind
- From simulating actions to simulating meaning
- Prediction as the unifying principle

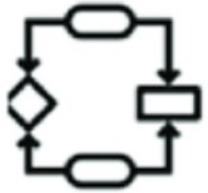


The Neuroscience of Language



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Summary

- Language as prediction, simulation, and structure
- Evolved from action, refined by culture
- Bridges biology, cognition, and meaning

Next Time

- Symbolic thought and human culture